Advance Python Assignment

Topic-3: Opening and Closing Files

Q1. Opening files in different modes in Python.

Ans.
In Python, files can be opened in different modes depending on the requirement. The open() function is used to open a file. Some common modes are 'r' for reading, 'w' for writing, and 'a' for appending data. The 'r+' and 'w+' modes allow both reading and writing. Each mode has its own purpose and is selected based on what operation we want to perform on the file. Using correct file mode is important to avoid data loss and errors while working with files.

Example:
```
file = open("sample.txt", "w")
file.write("Hello Python")
file.close()
```

Q2. Using open() and close() functions in Python.

Ans.
The open() function in Python is used to create or access a file, while the close() function is used to close the file after completing operations. When a file is opened, it stays active in memory, so it is important to close it properly to free resources. If we do not close the file, it may cause memory issues or data may not be saved correctly. Therefore, using open() and close() properly is very important in file handling.

Example:
```
file = open("data.txt", "w")
file.write("This is a file example")
file.close()
```